

Quick progress update

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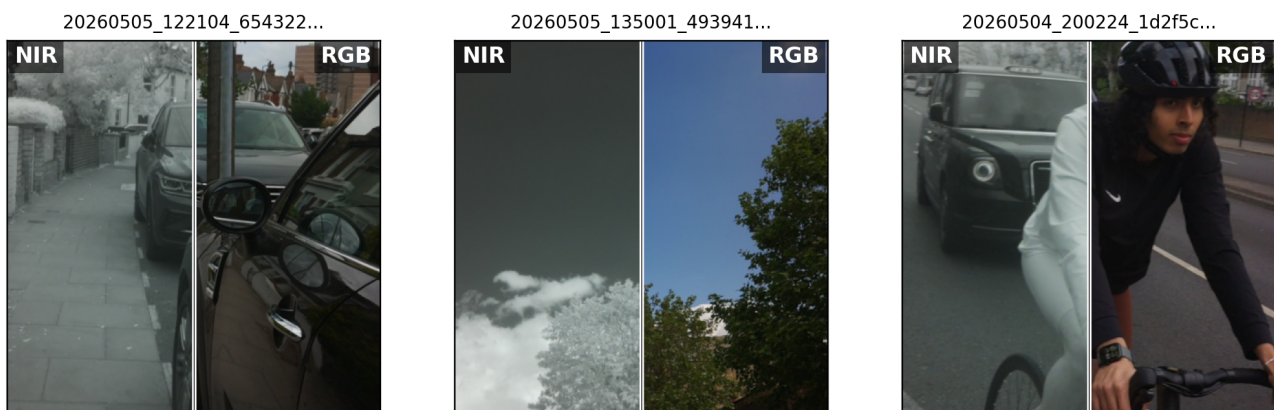
Dataset: paired and pixel-aligned

Collection since last update. Since the last update I have cycled for more than 20 hours around London collecting paired street-level RGB/NIR data. The capture device was mounted to my bag so it could record real scenes continuously while moving.



Current usable data: 12,536 paired RGB/NIR images after blur filtering and SIFT/RANSAC alignment. Split: 10,028 train / 1,253 validation / 1,255 test.

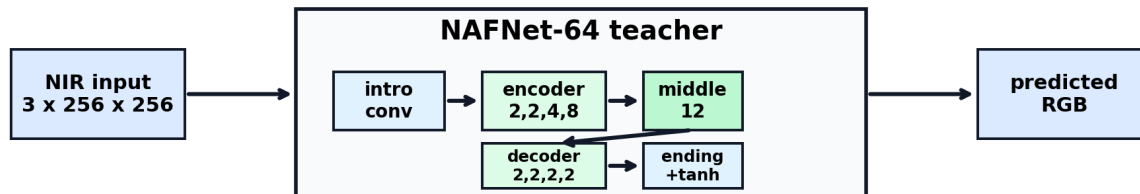
Pixel-paired training examples after blur filtering + SIFT/RANSAC alignment



Each tile is one aligned pair: left half NIR, right half RGB, same pixel grid.

Teacher model and downstream performance

Teacher architecture



Teacher: NAFNet width 64, 116.0M parameters

Training objective compares predicted RGB with ground-truth RGB:

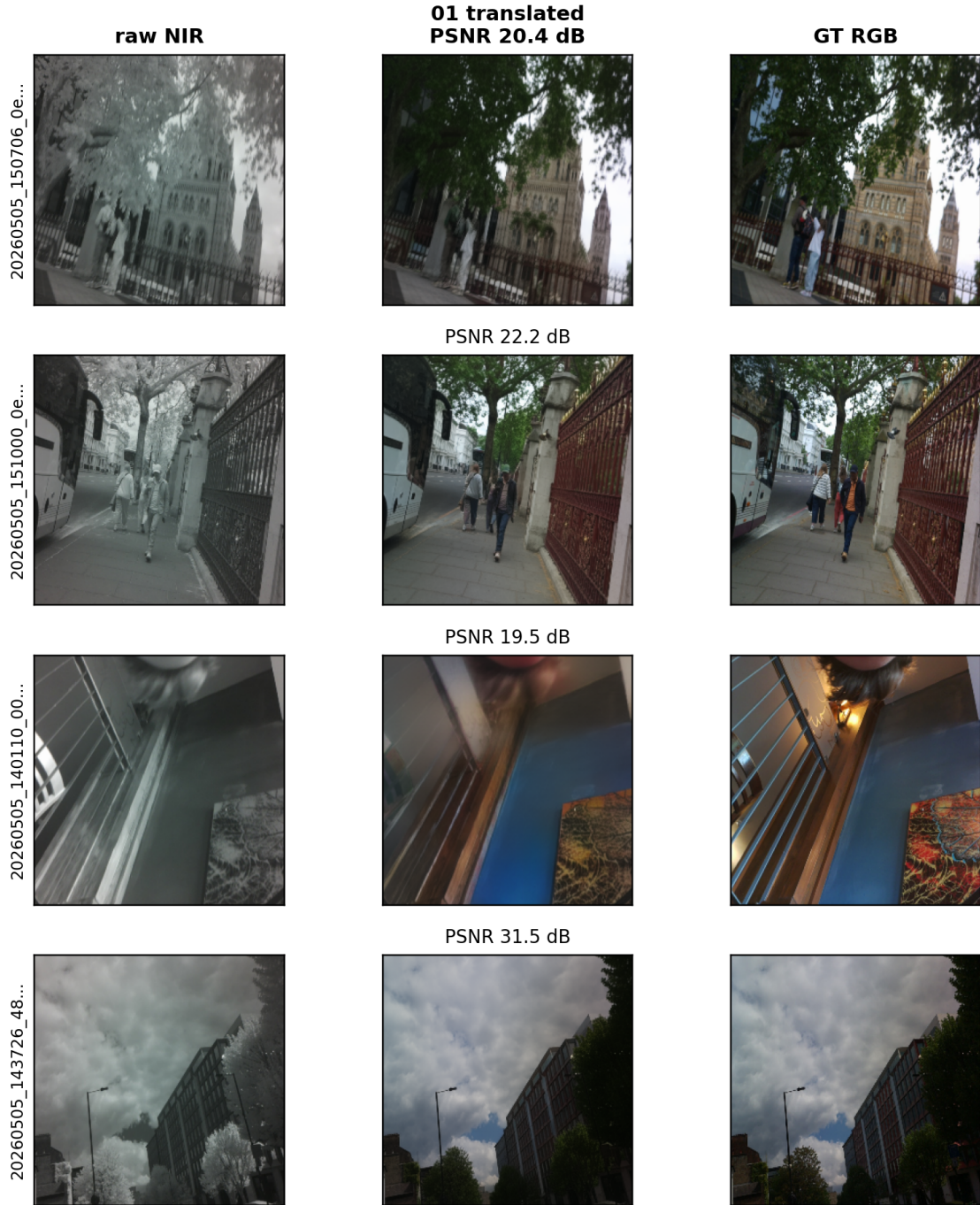
100 L1 + 20 VGG perceptual + 3 MS-SSIM + 0.5 GAN (PatchGAN)

I am using NAFNet for the teacher because it has worked best out of the architectures I have tried so far, and it looks like a reasonable architecture to distil into a smaller edge model later. NAFNet itself is not new; it is normally used for image restoration/reconstruction. As far as I can tell, it has not previously been used directly for NIR/IR/thermal to RGB image conversion. The useful part of this work is the paired dataset, applying NAFNet to this modality conversion problem, and testing whether it actually helps downstream RGB-trained models.

Model	Metric vs RGB	raw NIR	translated RGB	gap closure	result
MediaPipe hands	detection match	0.965	0.965	+0.0%	flat
MediaPipe selfie	mask IoU	0.887	0.789	-86.9%	worse
YOLO	matched IoU	0.778	0.797	+8.6%	better
DeepLabV3	mIoU	0.789	0.843	+25.4%	better
Mask R-CNN	matched IoU	0.761	0.784	+9.3%	better
ResNet50	top-1 agreement	0.295	0.530	+33.3%	better
MiDaS	depth rank corr.	0.863	0.954	+66.2%	better

The basic teacher model already improves most tested RGB-pretrained ML models versus raw NIR, especially MiDaS, ResNet50, DeepLabV3, YOLO, and Mask R-CNN. I am now looking at ways to make that gain bigger and less task-specific. It still does not fix MediaPipe-style skin/landmark tasks.

01 teacher test-set translations, mean PSNR 23.4 dB



Issues: Person examples can look plausible globally while local people/skin/clothing are wrong or muted. Likely reasons: people are often a small part of the 256×256 training crop, skin colour is weakly determined from NIR, and the pixel/perceptual losses are dominated by large background areas.

Attempts to improve downstream task performance

Smart crop

Smart crop is aimed at the person/MediaPipe failure mode. My idea is that when people are small in the frame, resizing the whole image to 256×256 leaves very little face/hand detail. This version uses cached MediaPipe boxes before resizing, so detected human/skin regions can be cropped first.

02 smart crop example: 20260505_143204_251574_f00ca076 (18.0% of frame kept before resize)



Region L1 boost

This targets the same issue without cropping. My idea is that faces/hands/people occupy too small a region, so normal L1 underweights them relative to pavement, sky, buildings, and trees. The model keeps the full frame but applies 5x L1 weight inside the cached MediaPipe region box.

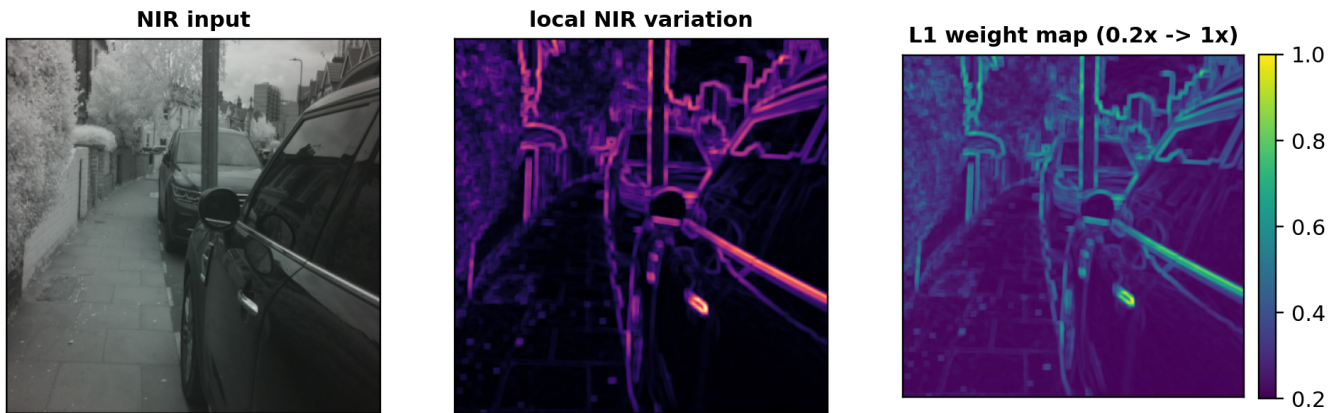
03 region-weighted L1: 20260505_150834_0e5f0f67 (18.0% of pixels boosted)



NIR-gradient L1

Some RGB colour is not inferable from flat NIR regions. My idea is that penalising those pixels equally can push the model toward average/desaturated colour, so this loss downweights L1 where local NIR variation is low.

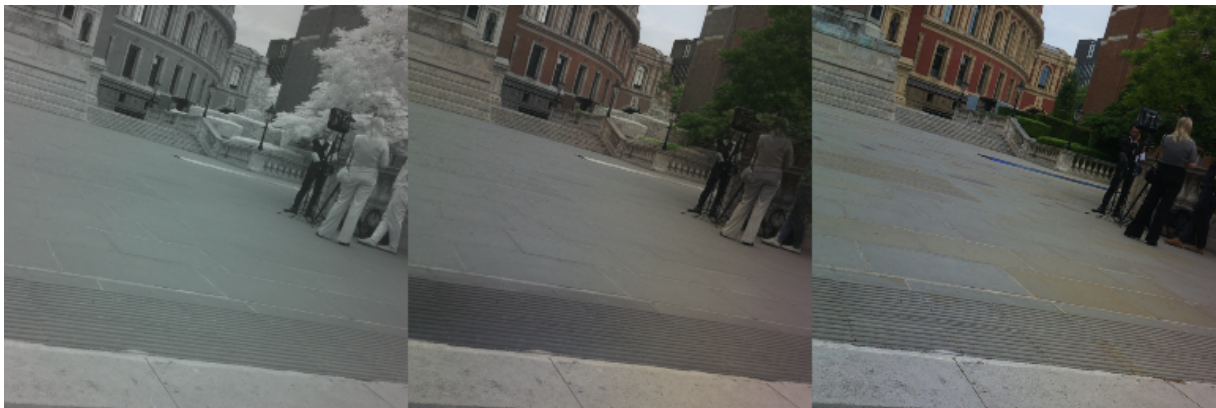
NIR-gradient L1: reduce penalty where NIR contains little predictive signal



Feature-map loss

My plan is to train the same teacher setup, but add a task-facing feature loss from frozen RGB-trained models. The generated RGB and real RGB are both passed through ResNet50 and DeepLabV3-ResNet50, and the generator is pushed to match their early/mid activations. The goal is to make the output useful to downstream RGB models, not just visually close under pixel losses.

This setup uses the baseline-style full frame, L_1 weight reduced to 25, and a feature-map weight ramp from 0 to 50 over the first 5 epochs.



All these approaches are at various stages, and I will update you with the results of each of these experiments next time we meet.

Training setup

I am currently using Google Cloud Platform with the \$300 free credit for training. Runs are split across a mixture of L4 and A100 spot instances: A100s for the heavier teacher/distillation jobs, and L4s for cheaper ablations, inference checks, and export/deployment work. Because spot instances can be interrupted, the training scripts save regular checkpoints and random sample outputs.

Deployment: distil the best teacher

The teacher is for quality. The deployment path is to distil into a smaller student and benchmark on edge hardware.

Distillation target: keep teacher behaviour, deploy small student



Distillation result. The test distillation run worked: the 6.0M-parameter NAFNet-24 student is learning the teacher output while staying close to ground-truth RGB, and the model is small enough to run on a Raspberry Pi 5.

Epoch	PSNR to GT	PSNR to teacher	SSIM to GT	SSIM to teacher
53	22.56 dB	27.02 dB	0.802	0.941

Student distillation progress, epoch 53: PSNR_GT 22.56 dB, PSNR_teacher 27.02 dB



Validation: SSIM_GT 0.802, SSIM_teacher 0.941. Student is learning to match the teacher while staying anchored to GT RGB.